

EDA Snacks – Implementation Plan

Machine Learning Project

1. Project Setup

- Define the project objective and target variable.
- Collect or import the snack dataset.
- Set up Python and required libraries.
- Suggested libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn.

2. Data Understanding

- Load the dataset and inspect records.
- Identify rows, columns, data types, and unique values.
- Generate descriptive statistics.

3. Data Cleaning

- Identify and handle missing values.
- Remove duplicate records where appropriate.
- Check invalid values such as negative prices or quantities.
- Standardize categorical values and column names.

4. Exploratory Data Analysis

- Perform univariate analysis of price, sales, ratings, discounts, and other numerical variables.
- Analyze category, brand, location, and other categorical variables.
- Use histograms, box plots, bar charts, and suitable visualizations.
- Study relationships such as price vs. sales, rating vs. sales, and category vs. sales.
- Generate a correlation matrix for relevant numerical variables.

5. Outlier Analysis

- Detect outliers using box plots and the IQR method.
- Investigate whether outliers are genuine observations or data errors.
- Remove, cap, transform, or retain outliers based on context.

6. Feature Engineering

- Create derived features such as revenue and discount percentage when appropriate.
- Extract month, weekday, or season if date information is available.
- Encode categorical variables for Machine Learning.
- Avoid features that would cause data leakage.

7. Feature Selection

- Analyze correlations and feature importance.
- Remove redundant or irrelevant variables.
- Separate predictor variables (X) and target variable (y).

8. Model Development

For sales prediction, candidate regression models include Linear Regression, Decision Tree Regressor, Random Forest Regressor, and Gradient Boosting. For demand classification, candidate models include Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting.

9. Model Training and Evaluation

- Split the data into training and testing sets.
- Train multiple candidate models.
- Compare models using appropriate evaluation metrics.
- Regression: MAE, RMSE, and R^2 .
- Classification: Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

10. Final Deliverables

Deliverable	Output
Cleaned dataset	ML-ready snack dataset
EDA	Statistics, visualizations, and findings
Features	Engineered and encoded variables
ML models	Trained candidate models
Evaluation	Performance metrics and comparison
Report	Methodology, results, and conclusion

11. Recommended Workflow

Raw Dataset → Data Understanding → Data Cleaning → EDA → Outlier Treatment → Feature Engineering → Feature Selection → Train/Test Split → Model Training → Evaluation → Final Report

12. Suggested Timeline

Week	Activities
1	Problem definition, dataset collection, project setup
2	Data understanding and cleaning
3	EDA and visualization
4	Outlier analysis and feature engineering

5	Feature selection and model development
6	Model evaluation, documentation, and final presentation